

BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE 340: Computer Architecture

Examination: Final
Duration: 1 Hour 50 Minutes

Semester: Summer 2026

Set: A
Full Marks: 50

Answer the following questions. Show calculations where required. Figures in the right margin indicate marks. Understanding the question is part of the examination.

Name:	ID:	Section:
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1. CO1

- a) In a RISC-V hackathon, you are required to transfer an IEEE 754 single-precision floating-point value from register F_7 to F_{14} . Direct transfer between the floating-point registers is prohibited; an integer register ($X_1 - X_{31}$) must be used as an intermediary. Determine whether this transfer is possible. If so, write the names of the required RISC-V instructions; otherwise, justify your answer. [1+2]
- b) Explain why IEEE 754 reserves the all-0s and all-1s exponent patterns for special cases instead of using them for ordinary normalized numbers? [3]
- c) Determine whether the following statements are True or false. If false, then write the correct answer.
 - i. We can only represent positive integers in the IEEE 754 floating point representation's biased exponent field. [1.5]
 - ii. Changing only the sign bit of a positive IEEE 754 number always changes its magnitude. [1.5]
- d) Variables a , b , d , and e contain IEEE 754 single-precision values and are mapped to registers F_4 , F_5 , F_6 , and F_7 , respectively. `arr` is a word array with base address stored in X_{20} . Convert the following high-level code into the corresponding RISC-V assembly code: [4]

<pre>if (a > b): a = -(a * d) - a elif (a == b): b = (a * d) - a else: arr[25] = b / e</pre>	Restrictions: Each expression $-(a * d) - a$ and $(a * d) - a$ must be evaluated using a single RISC-V floating-point instruction.
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2. CO2

- a) Explain why the immediate bits in RISC-V S-type instruction are divided into separate fields instead of being placed together? [3]
- b) Explain why RISC-V uses PC-relative addressing instead of the full target address? [3]
- c) Explain why left shift instruction cannot always replace a general multiplication instruction in RISC-V? [3]

3. CO3

- a) Consider the following instruction mix for a pipelined processor:

R-type: 20%	I-type(non-load): 15%	Load: 25%	Store: 15%	Branch: 25%
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- i) For what percentage of all instructions is the second ALU operand supplied by the immediate field? [2]
 - ii) For what percentage of all instructions is the value written back to the register file taken directly from the ALU output? [2]
 - iii) For what percentage of all instructions must the immediate field be sign-extended? [2]
- b) Design a single-cycle Datapath capable of executing **ONLY** the following instruction:

BEQ X5, X6, Else.

Note: Your diagram must label bit widths and the direction of data flow. Designing a general-purpose Datapath instead of one specific to this instruction will incur a severe penalty. [5]

Please Turn Over

- c) Analyze the tables given below. Table 1 assumes that individual stages of a **single-cycle datapath** have the following latencies; some values are missing.

Table 2 includes the total time required to execute some of the instructions using the given datapath.

Table - 1:

IF = 100ps	ID = ?	EX = 200ps	MEM = ?	WB = ?
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Table - 2:

Instruction	Time (PS)
BEQ $X_0, X_0, L1$	450
SD $X_{29}, 8(X_{20})$	750
LD $X_{26}, 0(X_{25})$	850

- Using the given data, find the time required for the ID, MEM, and WB stages. Show all the necessary calculations in the script. [2]
 - Suppose the same datapath is implemented as a 5-stage pipeline. What will be the clock period, and which stage determines it? [2]
 - A computer architect proposes modifying this pipelined datapath by splitting exactly one stage into two new stages, each taking exactly half of the original stage's latency. Which stage should be split to achieve the greatest possible reduction in clock period? Determine the new clock period after the split. [2]
- d) The code below runs on a **pipelined datapath** where the register X_{20} is initialized to $(10)_{10}$, and X_{21} is initialized to $(20)_{10}$. However, the programmer did not handle any of the data hazards in the code, so no stalling, forwarding, or code scheduling is applied.

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addi x20, x21, 4
add  x22, x20, x21
sub  x23, x20, x21
add  x24, x20, x20
```

- Determine the final values written into X_{22} , X_{23} , and X_{24} registers after execution. [2]
- For each of the three values, state whether it is the value the programmer intended, or a different value caused by a hazard. [2]

4. CO4

- a) Consider the following code sequence:

1. LUI X5, 0x10010 # base of array C 2. ADDI X6, X0, 200 # loop bound = 200 elements 3. ADDI X8, X0, 0 # i = 0 4. ADDI X9, X0, 0 # sum = 0 Loop: 5. LD X10, 0(X5) # load C[i] 6. ADD X9, X9, X10 # sum += C[i]	7. ADDI X5, X5, 8 # move to next C[i+1] 8. ADDI X8, X8, 1 9. BLT X8, X6, Loop
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- Which register in this loop best demonstrates temporal locality, and why? [2]
 - Which memory access in this loop best demonstrates spatial locality, and why? [2]
- b) Suppose the system has a 1.5 ns clock with a miss rate of 5%. It takes 2 clock cycles to determine if an access is a hit or a miss. It takes 1ns to pass the data from cache to CPU, and it takes 6ns to copy a block from the lower-level memory to cache. Determine the following: **Hit time, Miss Penalty, and AMAT**. [3]